

Introduction: The instructions below are a quick demo on how one would use the GLYZE UI from beginning to end, with the functionality it currently has. Once you complete the instructions below, please feel free to roam around and mess with its features. Let us know if there is anything you would like for us to change or could use improvement. Once again, thank you for your time!

Download Instructions: The GLYZE package, when received, will come in a .zip file. Please unzip the file in the terminal and `cd` into the directory. From there, run the command ‘`pip install .`’, where the python package manager will download the package locally. All done!

To load up the webpage, run the following command in the terminal: ‘`glyze`’

Step 1: When loading up the UI, you will be met with the home page that has links to all available pages. On the left is a bar that will also allow you to change between pages. Before using any of the pages, you must begin by clicking the Glyceride Mix page, where you can initialize your first mixture! Once you have done that, you should see:

The screenshot shows the GLYZE web application interface. On the left is a sidebar with navigation links: Home, Batch Reactor, Deodorizer, Glyceride Mix (highlighted), SFC Calculator, and Viscosity Model. The main area has a header with the GLYZE logo and a sub-header: 'Build and export a glyceride mixture — add by name or by structure, edit concentrations, switch units.' Below this is a large input field. The interface is divided into three main sections: 'Add species', 'Mixture', and 'Export'. The 'Add species' section has buttons for 'Built-in fatty acids', 'Built-in glycerides', 'Quick add extra species', and 'Quick add by name'. The 'Mixture' section has a 'Display/edit units' dropdown set to 'Moles', a 'Clear mix' button, and a table with two columns: 'Species' and 'Quantity'. The table contains two rows: '0 N05D00' with quantity '1' and '1 N06D00' with quantity '1'. Below the table is an 'Apply table edits' button. At the bottom of the 'Mixture' section, it shows 'Total moles (internal)' as 2 and 'Total mass (g)' as 218.293. The 'Export' section has an 'Export units' dropdown set to 'Moles', a 'Download CSV' button, and an 'Upload your CSV file' section with a 'Browse files' button and an 'Import CSV' button.

Species	Quantity
0 N05D00	1
1 N06D00	1

On the left is the column allowing you to add species to the mixture using our naming scheme, the middle allows you to view the mixture and change the units it is displayed in, and the right allows you to import previous mixtures or download the current mixture.

Once you are done, add 1 mole of caproic acid and 1 mole of oleic acid. For a note on the naming scheme, look at the note shown below.

Quick Note on the Naming Scheme:

Fatty Acids:

N06D00 - read as fatty acid with 6 carbons, no double bonds. Caproic acid

N18D01P09Z - read as fatty acid with 18 carbons and a double bond at position 9 with double bond class Z. Oleic acid

Glycerides:

G_N06D00_N06D00_N06D00 - read as the glyceride with 3 caproic acids attached to the glycerol backbone. Tricaproin.

G_N18D01P09Z_N18D01P09Z_EMPTY - read as the glyceride with 2 oleic acids attached to the glycerol backbone with an empty end.

You can mix and match all you like. This naming scheme gave the package an easy way to uniquely name all acids. In the future, we hope to add a bar with the most common species so that you don't have to continue writing these long names.

Step 2: Once you are finished, go to the batch reactor page. Here you will begin your first simulated reaction, esterification. Once you see your mixture as being correct, click *Initialize with Mixture*

The screenshot shows the 'Batch Reactor' interface. At the top, it says 'Initialize with a mixture, build reactions, then run the simulation.' Below this are two tabs: 'Esterification' (selected) and 'Interesterification'. A message states 'Loaded 2 species from the current mixture.' The section 'Current Glyceride Mix' contains a table with two columns: 'Species' and 'Moles'.

	Species	Moles
0	N06D00	1
1	N18D01P09Z	1

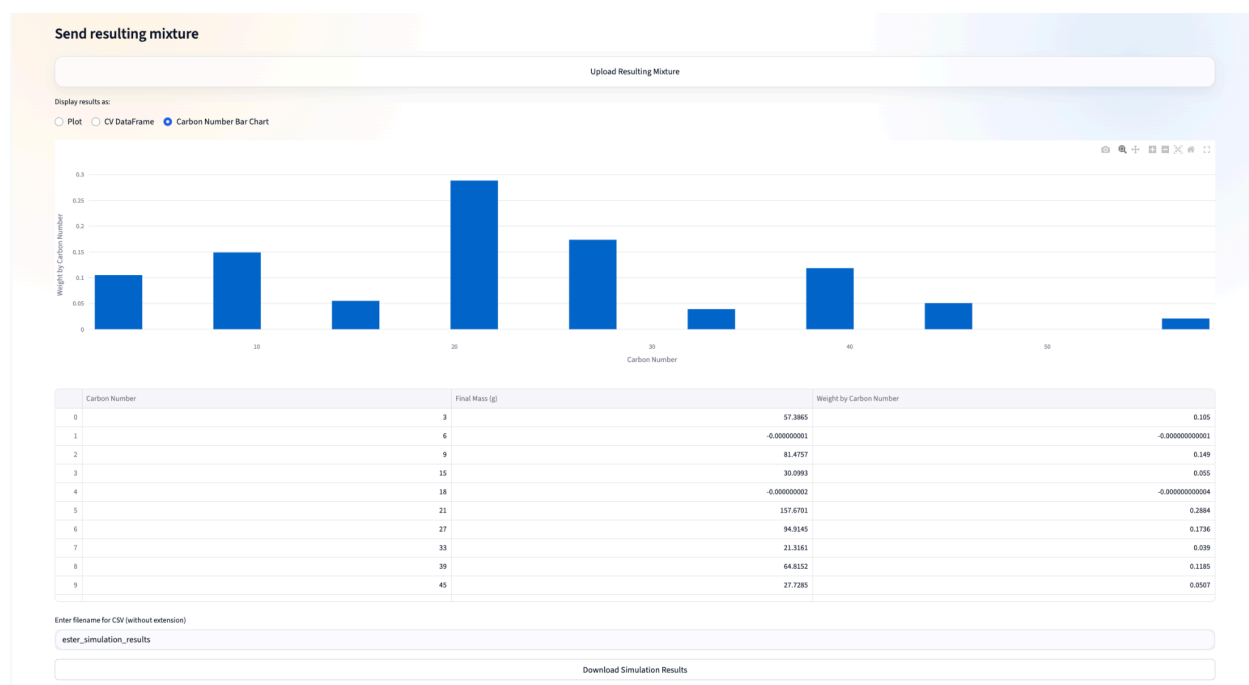
Below the table is a button labeled 'Initialize with Mixture'.

You will then see Feed Composition, where you will add how many moles of glycerol you want to add to the mixture. Put in 2 moles of glycerol for the time being.

Next, you will see a long list of reaction rate equations pop up, each with their rate constant k on the left. On the top is also an option to scale all rate constants by a particular amount. This allows you to find reaction rates if need be. For the time being, click *Initialize Simulation*

Next, you will be allowed to change simulation parameters. For the most part, you won't have to change these, but you may if you wish. For now, click *Run Simulation*.

Once you are done, you can view the results either with a plot, a CV, or a bar chart demonstrating what species with what carbon number are present in the current mixture.



If all is well, you can either download the simulation results with a CV and/or upload the resulting mixture, saving the results to be the new mixture.

Upload the resulting mixture.

Step 3: Once you are done, scroll to the top of the page and click Interesterification. If the current mixture looks correct, click *Initialize with Mixture*

Batch Reactor

Initialize with a mixture, build reactions, then run the simulation.

Esterification **Interesterification**

Loaded 19 species from the current mixture.

Current Glyceride Mix <>

Species	Moles
0 Glycerol	0.6231
1 G_EMPTY_N06D00_EMPTY	0.2466
2 G_EMPTY_EMPTY_N06D00	0.1817
3 G_N06D00_EMPTY_N06D00	0.0325
4 G_EMPTY_N06D00_N06D00	0.0719
5 G_EMPTY_N18D01P09Z_EMPTY	0.2466
6 G_EMPTY_EMPTY_N18D01P09Z	0.1817
7 G_EMPTY_N18D01P09Z_N06D00	0.0719
8 G_N06D00_EMPTY_N18D01P09Z	0.0649
9 G_EMPTY_N06D00_N18D01P09Z	0.0719

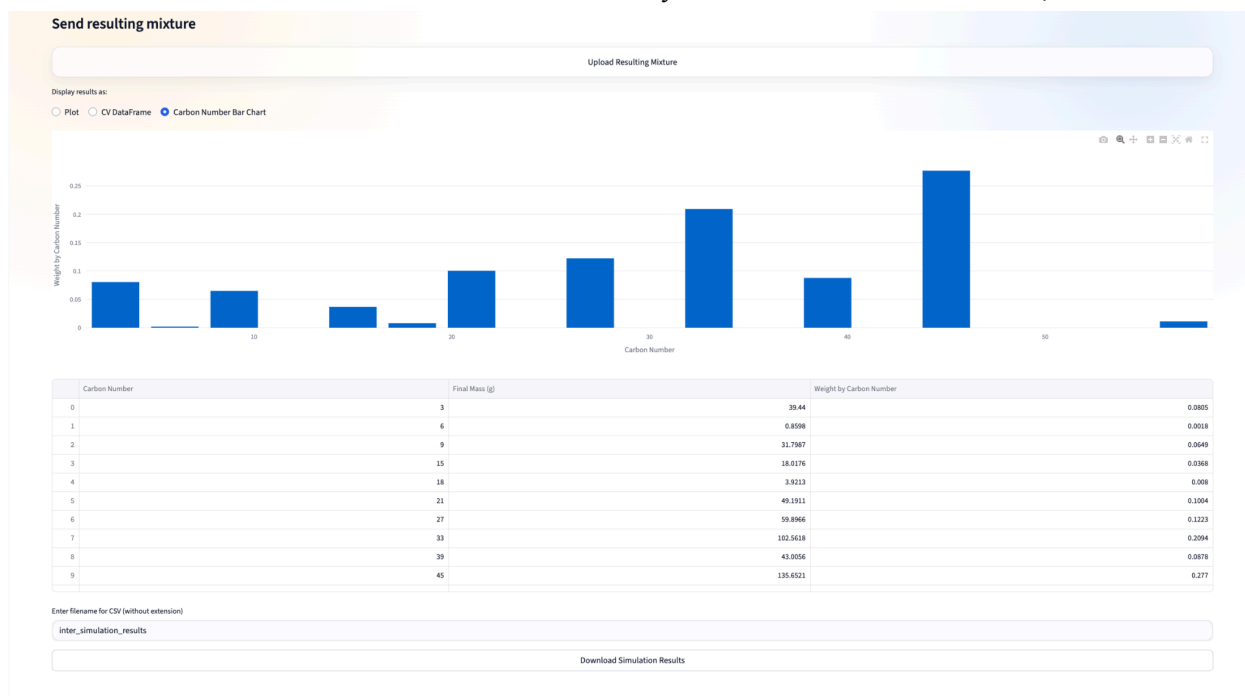
Initialize with Mixture

From here, a Plucked/Arranged list will show up that will allow you to take fatty acids off the ends or middles of species and put them back on the ends or middles of species respectively. You can alter them as you like, but for now [click Add Reactions](#)

As before, a list of all rate reaction equations present in the reaction will pop up. You can alter the rate constants as you like but for now, [click Initialize Simulation](#)

Finally, you can change the simulation parameters. For now, [click Run Simulation](#)

You can then view the results with the same functionality as the esterification simulation, as shown below.



Finally [click Upload Resulting Mixture](#)

Step 4: With our new mixture, we can now move on to the deodorization page. On this page, you can control various parameters such as the pressure, temperature, entrainment factor, and steam factor. For now, set the steam factor to 2. Then, [click Run Deodorizer](#). Once you are done, you can scroll down to the results.



Note that your results will look different from mine. You can look at your results either via table or bar chart. Once you are satisfied with your results, scroll down and [click Upload Resulting Mixture.](#)

This will save your results!

Step 5: we can now query the qualities that it has using some heuristics. For now, we can only test its viscosity, which is exactly what we will do. Load up the Viscosity Model page. You can keep all of the current parameters, but for now just [click Calculate Viscosity.](#)

From there, you will be greeted with a graph of the viscosity of the mixture!

In the future we hope to implement SFC. I hope you enjoyed GLYZE!